

How EGRESS Identifies Room Functions & Guarantees Occupancy Accuracy

Demystifying the Zero-Trust Architecture: How Semantic Room Names, Statutory Code Factors, and Physical Geometry Work Together

Core Question: How does EGRESS know room types without making dangerous assumptions?	Key Rule: Semantic Name Classification + Independent Geometry Math
Governing Authority: UAE Fire & Life Safety Code Table 3.13 (CDGH-OP-25, 2018)	Integrity Standard: Zero Tolerance for Unverified Drawing Occupant Numbers

1. The Essential Question & The Critical Distinction

A critical question often arises when evaluating automated compliance software:
'If EGRESS ignores drawing text notes, how does it know whether a room is an office, a meeting room, a cafeteria, or a storage room? If the software just assumes a generic number without reading proper room labels, won't the safety calculations be wrong?'

To understand why EGRESS is 100% accurate, we must make a fundamental engineering distinction between **Two Very Different Types of Drawing Text**:

Drawing Text Category	Example Text on Blueprint	How EGRESS Handles It	Why This Is Essential for Safety
1. Room Function Name (Architectural Tag)	'MEETING ROOM 1A' 'OPEN OFFICE WEST' 'SERVER / STORAGE' 'PANTRY / BREAKOUT'	WE READ & USE THIS. The engine reads the room name to identify the statutory occupancy category from UAE Code Table 3.13.	Ensures the correct density factor is selected (e.g. 1.4 m²/p for meetings vs. 9.3 m²/p for offices).
2. Pre-Written Occupant Count (Draftsperson Note)	'Occ: 79' 'Capacity: 10' 'Total: 250 persons' 'Max Load: 50'	WE REJECT & IGNORE THIS. The engine never trusts pre-calculated numbers written on the blueprint.	Prevents human calculation errors, copy-paste drafting mistakes, or deliberate manipulation to bypass safety rules.

The Core Principle in One Sentence

EGRESS **reads the room's function name** to determine the legal code density factor, but it **never trusts pre-written occupant counts**—it calculates the actual occupant count independently by dividing the exact geometric floor area (m²) by the code factor.

2. The 4-Tier Room Classification Algorithm

To ensure zero guesswork and absolute legal compliance, EGRESS evaluates every room polygon through a deterministic 4-tier classification pipeline:

Tier Level	Classification Stage	How It Operates	UAE Code Factor Applied
Tier 1 (Primary)	Semantic Keyword Tag Recognition	The engine reads the spatial text blocks located inside the room polygon boundary. It matches architectural keywords against the official UAE Code dictionary.	• 'MEETING' / 'PANTRY' → 1.4 m²/p • 'OFFICE' / 'CABIN' → 9.3 m²/p • 'WORKSTATION' → 4.6 m²/p • 'STORAGE' / 'SERVER' → 27.9 m²/p • 'RETAIL' / 'SHOP' → 2.8 m²/p
Tier 2 (Fallback)	Project Baseline Occupancy	If an architect left a room completely unlabeled (no text inside the walls), the engine applies the baseline building classification selected at upload (e.g. Commercial Business Office).	Applies standard building rate (e.g. 9.3 m²/p for business offices per UAE Code Table 3.13).
Tier 3 (Spatial)	Geometric & Egress Identification	Egress enclosures (fire stairs, smokeproof towers, elevator shafts) are identified by their exit door connectivity and layer metadata. Egress stairs generate 0 occupants.	Egress stairs = 0 occupants (they accommodate egress flow, not populate it).
Tier 4 (Override)	Interactive Consultant Override	In the interactive review dashboard, the engineer sees the assigned factor for every room. They can change any room's classification with one click (e.g. switching an office to a high-density call center), triggering instant recalculation.	Dynamic recalculation of all travel paths, door requirements, and corridor widths in real time.

3. Real-World Proof: How the Dubai Test Tower Rooms are Calculated

Below are the real calculations executed on the Dubai Commercial Building Typical Floor, proving how room name recognition and geometry work together:

Room Name Tag	Recognized Function	Polygon Area	Table 3.13 Factor	Independent Math Calculation	Final Load
OPEN OFFICE WEST	Regular Office	65.0 m²	9.3 m²/person	$65.0 \div 9.3 = 6.99 \rightarrow \text{ceil}(6.99)$	7 persons
OPEN OFFICE CENTRAL	Regular Office	118.0 m²	9.3 m²/person	$118.0 \div 9.3 = 12.69 \rightarrow \text{ceil}(12.69)$	13 persons
OPEN OFFICE EAST	Regular Office	65.0 m²	9.3 m²/person	$65.0 \div 9.3 = 6.99 \rightarrow \text{ceil}(6.99)$	7 persons
MEETING ROOM 1A	Assembly / Meeting	37.0 m²	1.4 m²/person	$37.0 \div 1.4 = 26.43 \rightarrow \text{ceil}(26.43)$	27 persons
MEETING ROOM 1B	Assembly / Meeting	37.0 m²	1.4 m²/person	$37.0 \div 1.4 = 26.43 \rightarrow \text{ceil}(26.43)$	27 persons
PANTRY / BREAKOUT	Assembly / Dining	37.0 m²	1.4 m²/person	$37.0 \div 1.4 = 26.43 \rightarrow \text{ceil}(26.43)$	27 persons
SERVER / STORAGE	Storage / Utility	82.0 m²	27.9 m²/person	$82.0 \div 27.9 = 2.94 \rightarrow \text{ceil}(2.94)$	3 persons
EXIT STAIR S-01	Egress Stairwell	12.5 m²	Egress Enclosure	Egress component — 0 load	0 persons

4. Why This Approach Eliminates Dangerous Life Safety Errors

By combining semantic room name reading with independent geometric math, EGRESS solves both failure modes:

- **Protection Against Mislabeling:** If an architect draws a 118 m² open office and mistakenly copies a note from a small room saying 'Occ: 4', EGRESS ignores the 4, computes $118 \text{ m}^2 \div 9.3 \text{ m}^2/\text{p} = 13$ occupants, and verifies life safety capacity accurately.
- **Protection Against Generic Assumptions:** EGRESS does not treat the whole floor as one big office. It knows that a 37 m² meeting room holds 27 people, while a 37 m² storage space holds only 2 people, applying the exact Table 3.13 density factor tailored to that space.
- **Ceiling Rounding Safety Standard:** Under international building codes, occupant load is always rounded UP to the next whole integer. A space calculating to 26.1 people is legally evaluated as 27 people, ensuring egress doors and stairs are never undersized.

5. FAQ: Questions from Auditors & Project Leadership

Q1: What if a floor plan has unusual room names like 'Zen Den' or 'Innovation Pod'?

A: EGRESS matches secondary keywords (e.g. 'Lounge', 'Breakout', 'Quiet', 'Work'). If no keyword matches, it assigns the baseline building office rate (9.3 m²/p) and highlights the space in the UI review workspace so the engineer can confirm or adjust the classification in one click.

Q2: What if a room is designed for higher density than standard code (e.g. a Call Center with compact desks)?

A: If an open office is designed at 4.6 m²/person (Concentrated Workstations) or 1.4 m²/person, the reviewer can select that specific occupancy subtype. The engine instantly recalculates the room's load (e.g. $118 \text{ m}^2 \div 1.4 = 85$ persons) and immediately checks if a second exit door is legally required.

Q3: Is this compliant with Dubai Civil Defence submission standards?

A: Yes. UAE Fire and Life Safety Code (CDGH-OP-25) Chapter 3, Section 4.1 explicitly states: *'Occupant load in any building or portion thereof shall be not less than the number of persons determined by dividing the floor area assigned to that use by the occupant load factor for that use as specified in Table 3.13.'* EGRESS implements this exact statutory formula.

Summary & Assurance

EGRESS does not guess, nor does it trust unverified draftsman numbers. It bridges the gap between architectural intent (room names) and physical reality (geometric floor area), delivering the most accurate, reliable, and legally defensible compliance audit in the industry.